Company name: IdeaForge Technology

One of India's leading manufacturers of UAS – unmanned aerial systems is IdeaForge Technology. It was founded in 2007. They cater to both the defence and the civilian markets. IdeaForge Technology has indigenously developed vertical take–off and landing drones. They are now one of the major suppliers to the armed forces, specifically the Indian Army and also the CAPF – central armed police forces, basically to the BSF – Border Security Force and to CRPF. They have completed over a million customer orders and are recognised as the world's leading dual-use drone in civil and also in the defence sector. After the conflict last year (Op Sindoor), the government has said they are starting an emergency drone procurement and has given around $2 billion in drone procurement green signal for indigenous drone manufacturers. The government has also asked for technical criteria such as electronic warfare EW resilience and sourcing restrictions on countries that don't share a border with India. Now, this segment, after this decision, has become highly competitive in the industry, with companies like IdeaForge, Garuda Aerospace, and Asteria Aerospace, and large conglomerates such as Adani Defence, Tata Advanced Systems, and L&T all competing for a small number of contracts.

Now, how should IdeaForge allocate R&D investments across various rounds where both rivals' capabilities and the government's own evaluation rules keep changing? This is where Game Theory is used.

IdeaForge's goal is not to reduce prices and win government contracts and deliver very basic tech; instead, they focus more on technology by investing heavily in Research and development, AI, autonomous navigation and secure drone systems. this how they use game theory to understand competitors' actions and make decisions according to that.

There are many defence drone manufacturing companies in India that are competing against each other in this demanding market.

The competitors include public sector players like HAL, BEL, and DRDO. Large private conglomerates like ADANI Defence, TATA Advanced Systems, and Larsen & Toubro. Special UAV firms like Ideaforge, Garuda Aerospace, Asteria Aerospace, Newspace Research, and Iotechworld. These companies come under the government of India's Atmanirbhar Bharat (self-reliant India) policy. This has production-linked incentives, an emergency procurement route with compressed timelines and, very importantly, the exclusion of Chinese-made components, all of which make this sector a highly competitive game

IdeaForge's product portfolio:

Netra – A surveillance mapping drone developed with DRDO's R&D; it is used for counter-jamming. Switch – it is a military UAV which is used for long-range, high endurance, high altitude last mile surveillance and security operations. Q series is used for public safety applications, border security, and mapping, and is also very useful during forest fires. And the newest toy is the ZOLT UAV; it is the eyes of operations, sees everything, never blinks an never stalls.

The companies are competing for:

- Long-term government contracts
- Surveillance drones for border areas
- Procurement contracts
- Export deals

Strategic problem

IdeaForge had a very important decision to make:

Should the company focus on investing heavily in research and development, building better AI-enabled drones, and maintaining premium pricing?

Or

Reduce the prices of their tech/product, compete with other companies to gain government contracts through cost leadership and put less investment in innovation.

The other competitors also face, to some extent, the same decision to make. The decision, therefore, was very crucial; Ideaforge did not just depend on its own strategy but also on competitors' choices.

Large conglomerates like Adani Defence, Tata Advanced Systems, L&T, and Garuda Aerospace, Asteria Aerospace, NewSpace Research; and the Government of India / Ministry of Defence, which acts simultaneously as customer, regulator and rule-setter.

The players are strategically interdependent for two reasons. First, each procurement round awards contracts to a limited set of winners, so one firm's chance of winning falls as rivals' capability rises. Second, the government's own evaluation criteria are partly shaped by what capabilities bidders demonstrate in trials, meaning firms' investment choices influence the very rules that will govern the next round. This dual interdependence between rivals, and between each firm and an evolving rule setter, is the classic condition under which Game Theory, rather than simple market-share analysis, becomes the more useful lens.

Game Theory Analysis.

Choosing the right model. procurement occurs in discrete, publicly observable rounds — emergency orders in June 2025, November 2025 and further rounds referenced by industry commentary as the "sixth round" of emergency procurement — and because firms can observe rivals' trial results and prior contract wins before the next round, this interaction is better modelled as a repeated, sequential game with a signaling element than as a single simultaneous-move game. ideaForge, as the longest-standing incumbent, has tended to move first by demonstrating new capability (for example, an EW-resilient ZOLT platform) ahead of a tender; rivals then decide whether to match. This gives ideaForge a potential first-mover (Stackelberg-type) advantage, reinforced by signaling: a publicly demonstrated "battle-tested" status is a credible signal to a government that cannot fully verify true capability in advance, and it raises the cost of imitation for rivals entering later.

How government policy shapes the competitive game each round.

Government policy and procurement trigger — Atmanirbhar Bharat creates preference for indigenous defence production; the emergency procurement route compresses tender timelines; the Ministry of Defence adds technical filters such as EW-resilience, secure communication, autonomous capability and restrictions on components sourced from non-border-nation suppliers. These rules raise the strategic value of proven domestic capability.

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Competitive field in each procurement round — Public-sector firms such as HAL and BEL bring institutional credibility; large conglomerates such as Adani Defence, Tata Advanced Systems and L&T bring capital, scale and manufacturing capacity; specialist UAV firms such as IdeaForge, Garuda Aerospace and Asteria Aerospace bring niche drone expertise. Since only a limited number of contracts are awarded, each firm’s payoff depends on both its own bid and the capability level revealed by rivals.

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IdeaForge’s strategic choice — IdeaForge can either follow a cost-leadership route by cutting prices and offering basic platforms, or follow a differentiation route by investing in R&D, AI-enabled autonomy, secure navigation, EW-resilient systems and strike-capable platforms. In game-theory terms, IdeaForge chooses a high-investment strategy to build a credible signal of quality before rivals can fully imitate it.

⇩

Rival response and strategic interdependence — Conglomerates may respond by scaling capital investment and using manufacturing depth to match technical specifications, while specialist peers may diversify into adjacent markets such as agriculture, logistics, mapping and EW-focused systems. Each rival observes IdeaForge’s demonstrations, trial performance and contract wins before choosing whether to match, undercut or differentiate in the next round.

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Game-theory outcome — The procurement market remains a repeated sequential game rather than a one-time price competition. IdeaForge gains a possible first-mover and signalling advantage if its technology is perceived as battle-tested and difficult to imitate. However, the advantage is not permanent because government criteria and rival capabilities evolve after every round. The likely equilibrium is a fragmented but fast-growing market in which IdeaForge maintains a credibility-based niche in tactical ISR and Mini UAVs, while larger firms compete on scale and specialist firms compete through focused innovation.

IdeaForge's round-level decision tree

| IdeaForges choice | |
|---|---|
| Invest heavily in next-gen R&D (EW-Resilient, strike-capable) | Maintain current ISR-focused capability |

| Rival Matches | Rival does not match | Rival matches | Rival does not match |
|---|---|---|---|
| Margins compressed for both,<br>Both stay Qualified(likely outcome) | IdeaForge gains share & signalling edge,<br>Rival lags behind on next round criteria. | IdeaForge risks exclusion if criteria tighten | Both remain in current niche,<br>low investment,<br>slower growth |

Payoff matrix: IdeaForge, Rival

| | Rival: Invest heavily | Rival: maintain current |
|---|---|---|
| IdeaForge: Invest heavily | (2,2) Nash Equilibrium | (4,1) |
| IdeaForge: Maintain Current | (1,4) | (3,3) Joint best Unstable |

For IdeaForge, Invest Heavily yields a better payoff than Maintain Current regardless of the rival's choice (2 beats 1 if the rival invests; 4 beats 3 if the rival does not) — the same logic applies symmetrically to the rival. Invest Heavily is therefore each player's dominant strategy, and (Invest Heavily, Invest Heavily) is the likely Nash equilibrium: neither player can improve its payoff by unilaterally switching strategy. Notably, this equilibrium (2,2) is worse for both firms than the mutual-restraint outcome (3,3) — a structure resembling a prisoner's dilemma, which is a common feature of R&D and capability "arms races" between competing suppliers to a single, criteria-setting buyer.

There are Two further dynamics that also matter a lot. First, reputational and signalling value compounds: each successful trial and delivery (such as SWITCH 2's induction into Army service) becomes a credible asset in the next round, serving as a form of entry deterrence against new conglomerate entrants lacking a comparable track record. Second, repetition also invites imitation — better-capitalised players observe which specifications win a round and can out-invest a mid-sized specialist in the next one. The available evidence does not resolve whether ideaForge's early-mover signalling advantage will persist as conglomerates scale up; this is best treated as an open strategic risk rather than a settled conclusion.

Strategic Implications:

IdeaForge appears to have gained: by investing early in EW resilience and extending beyond ISR into strike-oriented platforms, ideaForge has continued to win consecutive emergency-procurement awards through late 2025, suggesting its signalling strategy has, so far, preserved a favoured position in the tactical and mini-UAV segments even as reported revenue has been volatile.

Risks this strategy creates: heavy R&D and product-line expansion raise fixed costs at exactly the moments when revenue recognition is weak, as the 85% year-on-year revenue decline in Q1 FY26 illustrates. In a Prisoner's-Dilemma-type equilibrium, this cost pressure does not go away even if ideaForge wins the contract, because rivals investing at the same time compress industry-wide margins.

The competitor response: large diversified conglomerates entering with substantially deeper balance sheets could erode ideaForge's first-mover signalling advantage in later, larger rounds — particularly the roughly $2 billion tender reported for 2026 — where capital intensity may matter as much as UAV-specific engineering pedigree built up since 2009.

Sustainability: ideaForge's edge rests on accumulated trial credibility and product IP, which is a genuine and durable-looking asset, but one whose competitive value depends on the government continuing to reward demonstrated track record over sheer investment scale — a policy choice outside ideaForge's control.

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